Pulsed NMR of Protons in Liquids

Yonatan Haber, Yehonatan Tsitseluk[1]

[1] *Department of Physics, Faculty of Exact Sciences, Tel Aviv University*

June 2026

Pulsed proton NMR was used to characterise ten liquids: seven aqueous $CuSO_4$ solutions ranging from 0.125% to 8% by mass, distilled water, glycerine, and a $CuSO_4$ solution of unknown concentration. The longitudinal relaxation time T_1 was extracted by inversion recovery, the true transverse time T_2 by the CPMG spin-echo sequence, and the effective transverse time T_2^* from the free-induction decay. The proton gyromagnetic ratio was found to be approximately the same for all samples, matching the accepted value for protons in water once the thermal drift of the permanent magnet is accounted for. The expected hierarchy $T_1 \geq T_2 \geq T_2^*$ was confirmed for every sample. Both T_1 and T_2 were found to decrease with increasing ion concentration, while T_2^* was insensitive to it, since external-field inhomogeneity dominates the transverse dephasing. Glycerine, though free of paramagnetic ions, showed short relaxation times characteristic of slow molecular motion. From the linear dependence of the relaxation rates on concentration, the unknown solution was determined to be $(0.311 \pm 0.005)\%$.

1. INTRODUCTION & THEORY

Nuclear magnetic resonance (NMR) is based on the interaction between nuclear spins and a magnetic field. A nucleus with spin has a magnetic moment and behaves as a tiny magnetic dipole. When placed in a strong static field, these moments precess about the field at a characteristic rate called the Larmor frequency.

If an external radio-frequency (RF) field is then applied at that same frequency, it can tip the spins away from equilibrium. In pulsed NMR (PNMR) this is done with short RF pulses. Once a pulse ends, the spins relax back toward equilibrium through two processes: longitudinal relaxation, described by the time T_1, and transverse relaxation, described by T_2. These relaxation times depend on the molecular environment of the nuclei, so while the resonance frequency identifies the nucleus, the relaxation times identify the sample. This sensitivity of the relaxation times to the local environment is the physical basis for technologies such as magnetic resonance imaging (MRI).

1.1. Spin in a Magnetic Field

A nucleus with angular momentum $\vec{J}$ has a magnetic moment

$$\vec{\mu} = \gamma \vec{J} \tag{1}$$

where γ is the gyromagnetic ratio of the nuclear species.

In a static field $\vec{H}_0 = H_0 \hat{z}$ the moment experiences a torque equal to the rate of change of its angular momentum. With Eq. (1) this gives

$$\frac{d\vec{\mu}}{dt} = \gamma \vec{\mu} \times \vec{H}_0 \tag{2}$$

The change in $\vec{\mu}$ is perpendicular to both $\vec{\mu}$ and $\vec{H}_0$, so the moment precesses about the field at the Larmor frequency

$$\omega_0 = \gamma H_0 \tag{3}$$

The same result follows quantum mechanically from the Hamiltonian $\mathcal{H} = -\vec{\mu} \cdot \vec{H}_0 = -\gamma H_0 J_z$. For a spin-$\frac{1}{2}$ proton the two eigenstates are separated by

$$\Delta E = \gamma \hbar H_0 = \hbar \omega_0 \tag{4}$$

so a transition requires a photon of energy $\hbar\omega_0$, polarised in the plane perpendicular to $\vec{H}_0$. The two proton energy levels are shown in Fig. 1. For the proton the resonance lies in the radio-frequency range, $\nu_0[\text{MHz}] = 4.258\, H_0[\text{kG}]$ [1].

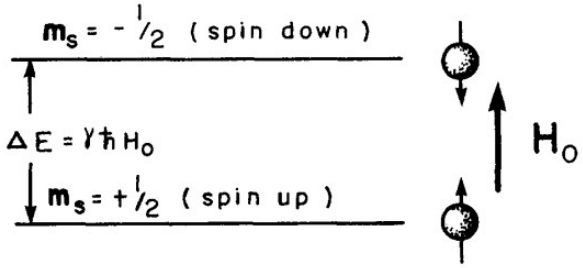


FIG. 1: Energy splitting of the two proton spin states in a static field H_0, separated by $\Delta E = \gamma \hbar H_0$. From Dixon [1].

1.2. Net Magnetisation

A single spin is not measurable. The observable quantity is the net magnetisation $\vec{M} = \sum_i \vec{\mu}_i$, whose ensemble average obeys the same equation of motion as a single moment,

$$\frac{d\vec{M}}{dt} = \gamma \vec{M} \times \vec{H}_0 \tag{5}$$

so $\vec{M}$ also precesses about $\vec{H}_0$ at ω_0.

At equilibrium the spin populations follow a Boltzmann distribution. For protons at room temperature the

Zeeman splitting satisfies $\Delta E \ll k_B T$, so the population difference is small and the equilibrium magnetisation, directed along the field, is

$$M_0 \approx N\mu \, \frac{\mu H_0}{k_B T} \tag{6}$$

with N the number of spins [1].

1.3. Motion under an RF Field

An oscillating field of angular frequency ω is applied transverse to $\vec{H}_0$ by a coil around the sample. A linear oscillation decomposes into two counter-rotating components, of which only the one co-rotating with the precessing magnetisation has a net effect, while the other averages to zero. In a frame rotating about $\hat{z}$ at ω, the precession due to H_0 is offset by an effective field $-\omega/\gamma$, and the equation of motion becomes

$$\left(\frac{d\vec{M}}{dt}\right)_{\text{rot}} = \gamma \, \vec{M} \times \left[\left(H_0 - \tfrac{\omega}{\gamma}\right)\hat{z} + H_1 \hat{x}'\right] \tag{7}$$

At resonance, $\omega = \omega_0 = \gamma H_0$, the longitudinal term vanishes and only H_1 along $\hat{x}'$ remains. The magnetisation then nutates about $\hat{x}'$ at the rate γH_1 while the pulse is applied, through an angle

$$\theta = \gamma H_1 \Delta t \tag{8}$$

for a pulse of duration Δt. The rotating frame and the nutation of $\vec{M}$ about H_1 are shown in Fig. 2.

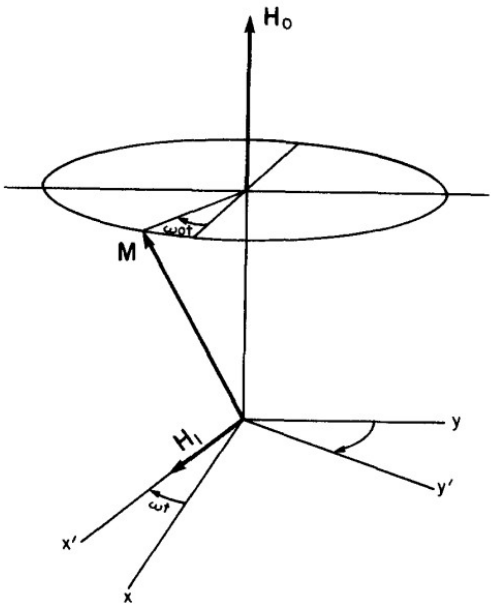


FIG. 2: Rotating-frame axes (x', y', z') with $z' = z$. The RF field H_1 is fixed along $\hat{x}'$, and at resonance $\vec{M}$ nutates about it in the y'–z' plane. From Dixon [1].

A pulse with $\theta = \pi/2$ (a 90° pulse) rotates the equilibrium magnetisation into the transverse plane, where its precession induces a decaying voltage in a pickup coil, the free-induction-decay (FID) signal. A pulse with $\theta = \pi$ (a 180° pulse) inverts the magnetisation to $-\hat{z}$ and produces no transverse signal on its own. A 90° phase shift of the RF changes the rotation axis from $\hat{x}'$ to $\hat{y}'$.

1.4. Relaxation

After a pulse, the magnetisation returns to equilibrium through two processes, with time constants T_1 and T_2.

1.4.1. Spin–Lattice Relaxation T_1

The longitudinal component recovers toward M_0 according to

$$\frac{dM_z}{dt} = -\frac{M_z - M_0}{T_1} \tag{9}$$

with solution an exponential approach to M_0 on the timescale T_1, the spin–lattice relaxation time. In particular, if the magnetisation is first inverted to $-M_0$ by a 180° pulse, the solution of Eq. (9) is

$$M_z(\tau) = M_0\left(1 - 2e^{-\tau/T_1}\right) \tag{10}$$

which crosses zero at $\tau_{\text{null}} = T_1 \ln 2$. Recovery requires energy exchange between the spins and the surrounding molecular motion, driven by fluctuating local fields with a spectral component at the Larmor frequency. Such fields arise from the dipole moments of neighbouring nuclei modulated by thermal motion, from molecular collisions, and from the unpaired electrons of paramagnetic ions, whose magnetic moment exceeds that of the proton by a factor of approximately 700 [1].

1.4.2. Transverse Relaxation T_2 and T_2^*

Spins tipped into the transverse plane start in phase but dephase over time, so the transverse magnetisation decays even without energy exchange. The dephasing of the spins in the rotating frame is shown in Fig. 3. The Bloch equation for the transverse components,

$$\frac{dM_{x,y}}{dt} = \gamma\,(\vec{M} \times \vec{H}_0)_{x,y} - \frac{M_{x,y}}{T_2} \tag{11}$$

introduces the spin–spin relaxation time T_2. Two mechanisms contribute to the dephasing. The first is the fluctuating internal fields, together with mutual spin flips that conserve the total z magnetisation. This constitutes the true T_2 process, and since every T_1 process also disturbs the transverse phase, $T_2 \le T_1$. The second is the static inhomogeneity of the external field. Spins in regions of different H_0 precess at different rates and fan out in phase. This contribution is reversible and is not a true relaxation process, but it adds to the observed FID decay. A single free-induction decay cannot separate the

two contributions, so what it measures is not T_2 but the combined effective transverse relaxation time T_2^*,

$$\frac{1}{T_2^*} = \frac{1}{T_2} + \gamma\,\Delta H \tag{12}$$

where ΔH is the field spread across the sample. Since the inhomogeneity term only adds to the rate, the hierarchy $T_2^* \le T_2 \le T_1$ holds for every sample, and recovering the true T_2 requires a sequence that reverses the static dephasing.

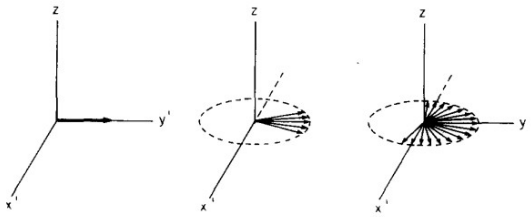


FIG. 3: Dephasing of the spins in the rotating frame due to differences in the local field, leading to decay of the transverse magnetisation.

1.4.3. Spin Echo and the CPMG Sequence

The inhomogeneity dephasing is static and can be reversed [2]. Following a 90° pulse the spins dephase for a time τ, and a 180° pulse then inverts the accumulated phases, so that faster and slower spins reconverge and form a spin echo at 2τ. This pulse sequence and the resulting echo are shown in Fig. 4. The reversible inhomogeneity is refocused at each echo while the irreversible T_2 processes are not, so a train of 180° pulses yields a sequence of echoes whose peak amplitudes decay as

$$V(t) = V_0\, e^{-t/T_2} \tag{13}$$

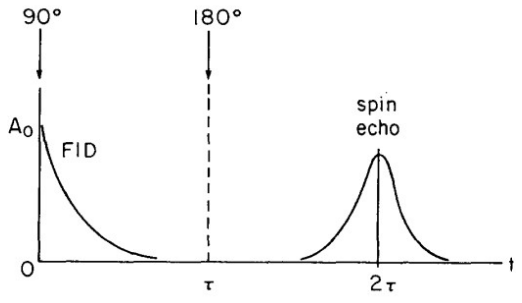


FIG. 4: Spin-echo sequence. After the 90° pulse the FID decays, and a 180° pulse applied at time τ produces a refocused echo at 2τ.

Applying the refocusing pulses about $\hat{y}'$ rather than $\hat{x}'$ defines the Carr–Purcell–Meiboom–Gill (CPMG) sequence. A 180° pulse tips the magnetisation slightly out of the transverse plane. With the CPMG phasing this error reverses on the following pulse, so it cancels on the even-numbered echoes instead of accumulating, and the even-echo envelope provides the true T_2 [1].

1.5. Relaxation Mechanisms in Liquids

1.5.1. Paramagnetic Ions

Dissolved paramagnetic ions such as Cu^{2+} from $CuSO_4$ carry an unpaired electron and produce strong fluctuating fields at neighbouring protons, which increases the relaxation rate [1]. In the dilute limit each ion acts independently, so both relaxation rates are linear in the ion concentration n [3],

$$\frac{1}{T_i} = \frac{1}{T_{i,0}} + r_i\, n \qquad i = 1, 2 \tag{14}$$

where $T_{i,0}$ is the relaxation time of the undoped solvent and r_i the ionic relaxivity. This relation provides a calibration from which the concentration of an unknown solution can be determined.

The electronic Larmor frequency lies far above the proton frequency, so the RF pulses do not excite the electrons and the measured signal comes from the protons alone. Oxygen nuclei contribute no signal, as they have no magnetic moment. Dissolved molecular O_2, however, is itself paramagnetic, so even nominally pure water in equilibrium with air is weakly doped and relaxes faster than degassed water [1].

1.5.2. Molecular Motion and Viscosity

The efficiency of all these fluctuating-field mechanisms is governed by the timescale of the molecular motion that produces them. In the fast-motion regime characteristic of non-viscous liquids the two relaxation processes are equally efficient and $T_1 \approx T_2$, while slower motion, richer in low-frequency spectral content, feeds the transverse dephasing preferentially and opens a gap between the two times [1]. The ratio T_1/T_2 is therefore a measure of how slowly the molecules move. It sits near unity for fast-tumbling liquids and grows well beyond it as the motion slows. Since the timescale of the motion grows with viscosity, a viscous but ion-free liquid such as glycerine is expected both to relax quickly and to show a pronounced T_1/T_2 separation.

1.6. Objectives

The objectives of the experiment are:

- To determine the proton gyromagnetic ratio γ from the Larmor frequency measured in a known field H_0 for every sample, to verify that the value is consistent across samples as expected for a property of the proton, and to compare it with the accepted value.

- To measure T_1, T_2 and T_2^* for a series of $CuSO_4$ solutions of decreasing concentration and for glycerine, to examine their dependence on ion concentration and on viscosity, and to verify that the expected hierarchy $T_1 \geq T_2 \geq T_2^*$ holds.
- To determine the concentration of an unknown $CuSO_4$ solution from the linear relation (14) between the relaxation rates and the ion concentration.

2. EXPERIMENTAL SYSTEM

The measurements were performed with the TeachSpin PSA-1 pulsed-NMR spectrometer [4], designed for the study of PNMR in liquid samples held in tubes. The sample sits in the bore of a permanent magnet, is excited by RF pulses through a transmitter coil, and its response is detected by a receiver coil, processed, and displayed on an oscilloscope connected to a computer for data export. The modules and signal flow are shown in Fig. 5, and the setup as installed in the laboratory in Fig. 6.

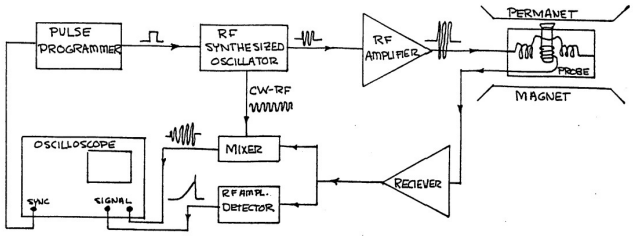


FIG. 5: Block diagram of the TeachSpin PSA-1 system.

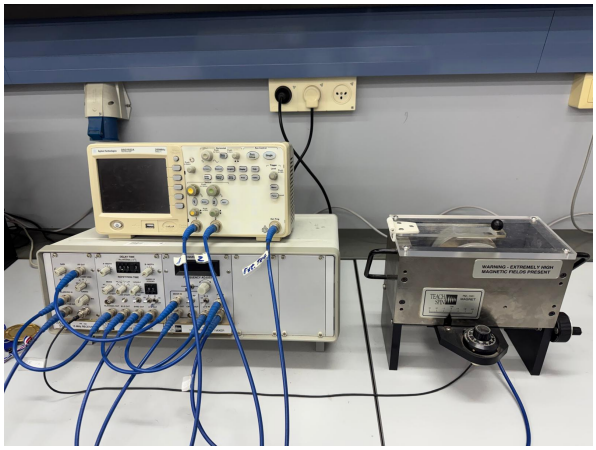


FIG. 6: The TeachSpin PSA-1 setup in the laboratory. On the left, the pulse programmer, RF synthesiser, receiver and mixer modules, with the oscilloscope on top. On the right, the permanent magnet in its shielded enclosure, with the sample-tube slot on top.

2.1. Permanent Magnet and Hall Probe

The static field H_0 is produced by a permanent magnet, between whose poles the sample tube is positioned. The field at the sample drifts with temperature at a documented rate of approximately 4 G/°C relative to the nominal 20 °C specification [4], so the laboratory temperature is recorded during the measurements.

2.2. Transmitter and Receiver Coils

Two coils surround the sample tube in perpendicular orientations. The transmitter coil, wound in a Helmholtz configuration, applies the RF field H_1 in the plane transverse to H_0. The receiver coil, mounted at right angles to the transmitter, detects the changing flux from the precessing magnetisation, which induces the EMF recorded as the FID signal. The perpendicular geometry reduces direct pickup of the transmitted pulse.

2.3. Pulse Programmer and RF Synthesiser

The pulse programmer generates the timed sequences from two pulse types, an A pulse sent first and a B pulse that may follow between one and 99 times. The width of each pulse is adjustable, which is how the 90° and 180° pulses are set. The delay τ between the A and B pulses is selectable, and for a train of B pulses the spacing is 2τ. A repetition time between successive firings of the sequence is also set. It must exceed the sequence length and, for quantitative measurements, allow the spin system to return to equilibrium. The B pulses can be phase-shifted to lie along $\hat{y}'$, as required for the CPMG sequence.

The RF synthesiser converts the programmer's pulses into radio-frequency bursts. Its frequency is tunable around 15.5 MHz with a resolution of 10 Hz, allowing it to be matched to the proton Larmor frequency. The output is amplified and sent to the transmitter coil, and a copy is routed to the mixer as a reference.

2.4. Receiver, Mixer and Acquisition

The receiver amplifies the EMF from the receiver coil and outputs a rectified envelope of the FID. Its gain and an adjustable RC time constant were held fixed to keep measurements comparable, with the RC setting also determining the recovery time the receiver needs after a transmit pulse before reporting the true signal. To protect the detector, the receiver coil is blanked during transmission, since the RF pulse would otherwise saturate it. The receiver response is also non-linear below about 2.5 V, so measurements were kept above this level where possible [4].

The mixer multiplies the receiver signal by the synthesiser reference and outputs the difference of their frequencies. When the reference matches the Larmor frequency the difference vanishes and the mixer output stops oscillating, which is used to locate resonance. The oscilloscope displays the FID on one channel and the mixer

output on the other, so the decay envelope and the resonance condition are observed together and rechecked as the field drifts. Each transient is then transferred from the oscilloscope to a computer for further analysis.

3. EXPERIMENTAL METHODS

3.1. Pulse Calibration and Repetition Timing

Three calibrations were performed at the start of each session. Firstly, the synthesiser was tuned to the proton Larmor frequency by adjusting it until the mixer output stopped oscillating, indicating a zero beat between the reference and the precession signal. Secondly, since the FID amplitude scales with $\sin\theta$ of the tip angle, the 90° pulse width was set by maximising the initial FID amplitude. Thirdly, the 180° pulse width was set by minimising the same amplitude, rather than nulling it, since the finite spatial spread of H_1 across the sample gives a small residual signal at the ideal $\theta = \pi$.

3.2. Gyromagnetic Ratio γ

The static field at the sample position was read directly from the digital Hall probe, whose 10 G resolution was treated as a uniform-distribution error giving $\sigma_{H_0} = 10/\sqrt{12} \approx 2.9$ G (Appendix 1). The Larmor frequency was located on the mixer channel. Following a single 90° pulse the mixer output beats at the difference between the synthesiser reference and the precession frequency, and the reference was tuned until the beat vanished, at which point it equals ν_0. This zero-beat tuning was performed twice independently for each sample, the mean adopted as ν_0 and half the difference as σ_{ν_0} (Appendix 2).

The gyromagnetic ratio follows from the resonance condition (3),

$$\gamma = \frac{2\pi\nu_0}{H_0} \tag{15}$$

with the two contributions combined in quadrature,

$$\sigma_\gamma = \gamma\sqrt{\left(\frac{\sigma_{\nu_0}}{\nu_0}\right)^2 + \left(\frac{\sigma_{H_0}}{H_0}\right)^2} \tag{16}$$

For the comparison with the accepted value, the field was corrected for the thermal drift of the permanent magnet at the measured laboratory temperature T,

$$H_{\mathrm{eff}} = H_0 + (T - 20\,^\circ\mathrm{C}) \times 4\ \mathrm{G}/^\circ\mathrm{C} \tag{17}$$

with the coefficient taken from the instrument manual [4] and treated as exact. The uncertainty of the thermometer reading contributes negligibly. As the reference, the resonance relation for protons in water quoted by Ref. [1], $\nu_0[\mathrm{MHz}] = 4.258\,H_0[\mathrm{kG}]$, is adopted. Through the Larmor relation (3)

$$\gamma_p = 26754 \pm 3\ \frac{\mathrm{rad}}{\mathrm{s}\cdot\mathrm{G}} \tag{18}$$

where the uncertainty follows from the four-significant-figure rounding of the quoted constant, an interval of ± 0.0005 in the factor 4.258.

3.3. Spin–Lattice Relaxation Time T_1

T_1 was measured with the inversion-recovery sequence 180°–τ–90°. The 180° pulse inverts the magnetisation, and after the variable delay τ the 90° pulse rotates the instantaneous $M_z(\tau)$ into the transverse plane, where the initial FID height is proportional to it. Because the receiver reports a rectified envelope, the sign of each amplitude was restored by hand about the observed null point. The signed peak amplitude, which fluctuates from firing to firing and was therefore assigned half its observed spread as uncertainty (Appendix 2), was recorded for roughly fifteen delays, spaced to span the full recovery from inversion to saturation, and fit to the form of Eq. (10),

$$V(\tau) = V_0\left(1 - 2e^{-\tau/T_1}\right) \tag{19}$$

with V_0 and T_1 free.

3.4. Effective Transverse Relaxation Time T_2^*

T_2^* was obtained from the FID following a single 90° pulse. Because the receiver requires a finite recovery (capacitor-charging) time after the transmit pulse, the first $\sim$25 μs of each trace were discarded and the remaining rise was modelled explicitly, the full FID being fit to

$$V(t) = V_0\left(1 - e^{-t/RC}\right)e^{-t/T_2^*} \tag{20}$$

with V_0, RC and T_2^* free, and each trace point weighted by the RMS noise of the pre-pulse baseline as its uncertainty (Appendix 3). By Eq. (12) the inhomogeneity term $\gamma\,\Delta H$ is expected to dominate the measured rate for all but the most concentrated solutions, so T_2^* should be essentially independent of concentration.

3.5. Spin–Spin Relaxation Time T_2

The true T_2 was measured with the CPMG sequence,

$$90^\circ_x - \tau - \left[\,180^\circ_y - 2\tau\,\right]_n \tag{21}$$

which produces spin echoes at times $t_k = 2k\tau$ whose even-numbered peaks decay according to Eq. (13). The pulse spacing τ and the number of echoes n were matched to each sample so that the train spanned several T_2, except for distilled water, where the repetition-time and field-drift constraints limited the accessible window to a fraction of T_2. The peak amplitude of each even echo was extracted numerically from the digitised trace and the envelope fit to Eq. (13), with the same baseline-noise uncertainty per point as for the FID.

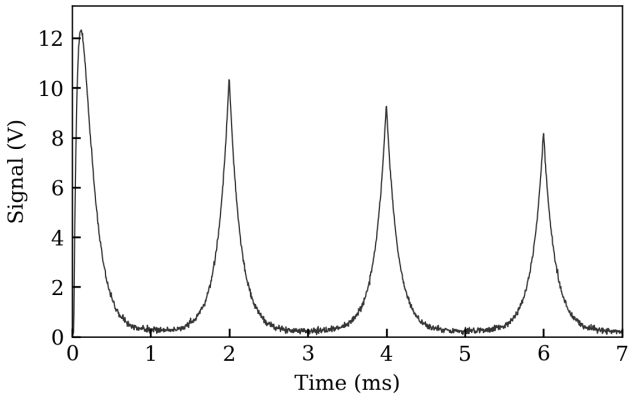


FIG. 7: Representative raw CPMG echo train for the 1% solution, showing the initial FID and the first three echoes. The peak amplitude of each even echo was extracted and fit to Eq. (13) to obtain T_2.

3.6. Concentration of the Unknown Solution

The concentration of the unknown $CuSO_4$ solution was determined by calibration against the known solutions through the linear relation (14). Plotting $1/T_1$ and $1/T_2$ against n for the known solutions and fitting straight lines (slope $m_i = r_i$, intercept $b_i = 1/T_{i,0}$) yields two calibration curves. Pure water was excluded from the fits because its uncertainty is an order of magnitude smaller than that of any doped sample, so under inverse-variance weighting it would dominate the regression and pull the line away from the concentration range of interest. The unknown concentration was recovered by projecting its measured rate onto each line,

$$n_{\text{unknown}} = \frac{1/T_{i,\text{unknown}} - b_i}{m_i} \tag{22}$$

with the uncertainty propagated through the fitted parameters and their covariance as detailed in Appendix 5. The two independent estimates were combined by an inverse-variance weighted mean, and their mutual consistency checked through N_σ.

4. RESULTS

Throughout the experiment the Hall probe read a constant field of $H_0 = 3570 \pm 3$ G at a laboratory temperature of 26.3 °C, with the zero-beat frequencies of all samples clustering near 15.29 MHz.

4.1. Gyromagnetic Ratio γ

The gyromagnetic ratio of each sample was obtained from Eq. (15) and is listed in Table I.

TABLE I: Gyromagnetic ratio γ for each sample. Uncertainties follow Eq. (16) and correspond to relative uncertainties of 0.09–0.10%, dominated by the common field term.

Solution	$\gamma \left[\frac{\text{rad}}{\text{s}\cdot\text{G}}\right]$
8%	26921 ± 25
4%	26917 ± 26
2%	26924 ± 25
1%	26911 ± 25
0.5%	26908 ± 26
0.25%	26913 ± 26
0.125%	26909 ± 27
H_2O	26915 ± 26
Glycerine	26931 ± 25
Unknown	26891 ± 25

The field error is common to all rows of Table I and cancels in comparisons between samples, which are therefore governed by the independent frequency scatter of about ± 12 rad s^{-1} G^{-1} per sample. On this scale the largest disagreement, glycerine against the unknown solution, is $N_\sigma \approx 2.4$, an unsurprising largest deviation among ten measurements, and γ is treated as a single quantity shared by all samples. Averaging gives

$$\gamma_{\text{avg}} = 26914 \pm 25 \ \frac{\text{rad}}{\text{s}\cdot\text{G}} \tag{23}$$

This result exceeds the accepted value γ_p (18) by 0.60%, or about 6.4 combined standard deviations. The discrepancy is consistent with a single documented systematic effect. At the measured 26.3 °C the thermal coefficient of Eq. (17) raises the effective field to $H_{\text{eff}} \approx 3595$ G, and since $\gamma \propto 1/H$ the corrected value is

$$\gamma_{\text{corr}} = 26726 \pm 25 \ \frac{\text{rad}}{\text{s}\cdot\text{G}} \tag{24}$$

which agrees with γ_p within $N_\sigma \approx 1.1$.

4.2. Spin–Lattice Relaxation Time T_1

For each sample T_1 was extracted by fitting the signed inversion-recovery amplitudes to Eq. (19), with the fitted T_1 using errors from the least-squares covariance matrix (Appendix 4). The overlaid fits are shown in Fig. 8 and the fitted values in Table II.

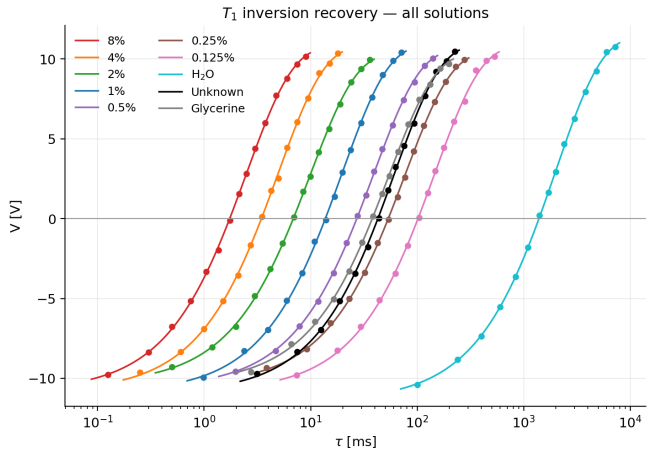


FIG. 8: Inversion-recovery voltage V versus delay time τ for all samples (log scale on the τ-axis). Solid lines are fits to Eq. (19).

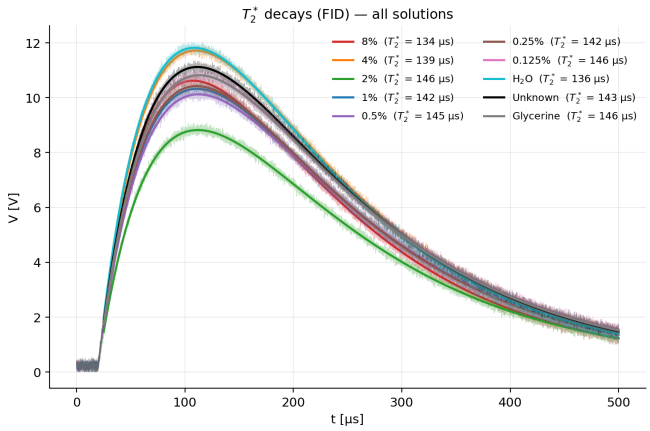


FIG. 9: Free-induction decays for all samples. Solid lines are fits to Eq. (20). Note the μs timescale, three orders of magnitude below those of Figs. 8 and 10.

TABLE II: Spin–lattice relaxation time T_1 for each sample. Values in parentheses are relative uncertainties (%).

Solution	T_1 [ms]	χ^2_{red}
8%	2.506 ± 0.014 (0.56%)	1.101
4%	4.973 ± 0.034 (0.69%)	1.116
2%	9.906 ± 0.053 (0.54%)	0.908
1%	19.81 ± 0.12 (0.58%)	1.121
0.5%	39.08 ± 0.17 (0.43%)	0.914
0.25%	77.18 ± 0.37 (0.47%)	0.939
0.125%	146.8 ± 1.1 (0.73%)	1.209
H_2O	2004 ± 8 (0.40%)	1.106
Glycerine	54.56 ± 0.34 (0.63%)	0.906
Unknown	61.67 ± 0.40 (0.65%)	0.914

T_1 increases monotonically as the $CuSO_4$ concentration decreases, spanning a factor of about 800 between the 8% solution and distilled water, with each halving of the concentration approximately doubling T_1, as anticipated by Eq. (14) in the ion-dominated regime. All χ^2_{red} values lie between 0.9 and 1.21, confirming both that the assigned amplitude uncertainties describe the scatter well and that the single-exponential recovery is the correct model. Glycerine, without paramagnetic ions, relaxes faster than any solution at 0.25% and below, its T_1 falling between those of the 0.5% and 0.25% samples, a point to be re-mentioned in the discussion.

4.3. Effective Transverse Relaxation Time T_2^*

T_2^* was obtained by fitting each FID to Eq. (20), with the fitted T_2^* using errors from the least-squares covariance matrix (Appendix 4). The envelopes are shown together in Fig. 9 and the fit results in Table III.

TABLE III: Effective transverse relaxation time T_2^* for each sample. All fit uncertainties are ± 1 μs (about 0.7%).

Solution	T_2^* [μs]	χ^2_{red}
8%	134 ± 1	0.961
4%	139 ± 1	1.112
2%	146 ± 1	0.994
1%	142 ± 1	1.217
0.5%	145 ± 1	1.122
0.25%	142 ± 1	0.928
0.125%	146 ± 1	1.028
H_2O	136 ± 1	0.980
Glycerine	146 ± 1	0.949
Unknown	143 ± 1	0.984

In sharp contrast to the 800-fold span of T_1, all T_2^* values fall within a band narrower than 10%, confirming that the FID decay is dominated by the static-inhomogeneity term of Eq. (12) rather than by the sample. The observed rate is the sum of the sample's intrinsic transverse rate and a fixed inhomogeneity contribution, and its near-constancy across samples means the latter dominates.

The FID can therefore be inverted into a measurement of the magnet itself. Here ΔH denotes the width (FWHM) of the distribution of static-field values across the sample volume between the magnet poles. Evaluating $\Delta H = (1/T_2^* - 1/T_2)/\gamma$ sample by sample, with the measured T_2 of Sec. 4.4, gives $\Delta H = 0.25$–0.27 G for every liquid. As a fraction of the main field H_0 this is about 72 parts per million (ppm), so the field is uniform to a few tens of ppm across the sample. The spread is small in absolute terms, yet still far larger than the sample's own $1/T_2$, which is why it dominates the FID decay.

4.4. Spin–Spin Relaxation Time T_2

The true T_2 was measured with the CPMG sequence, the even-echo peaks being fit to Eq. (13), with the fitted T_2 using errors from the least-squares covariance matrix (Appendix 4). The overlaid decays are shown in Fig. 10 and the results in Table IV.

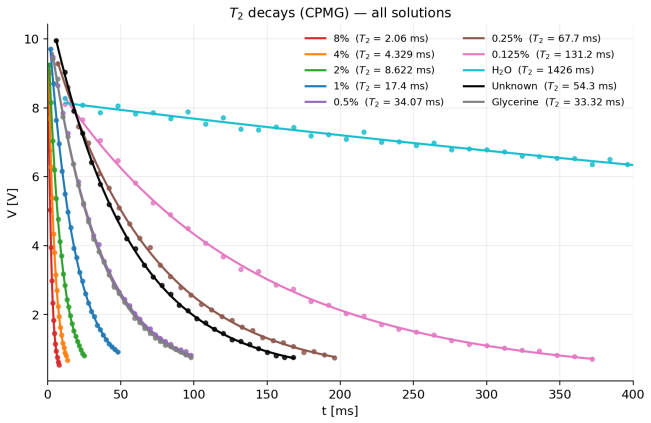


FIG. 10: CPMG spin-echo decays for all samples. Solid lines are single-exponential fits to Eq. (13), with the fitted T_2 values shown in the legend.

TABLE IV: Spin–spin relaxation time T_2 measured by the CPMG method. Values in parentheses are relative uncertainties (%). Where $\chi^2_{\mathrm{red}} \gg 1$ the quoted uncertainties are lower bounds, as discussed below.

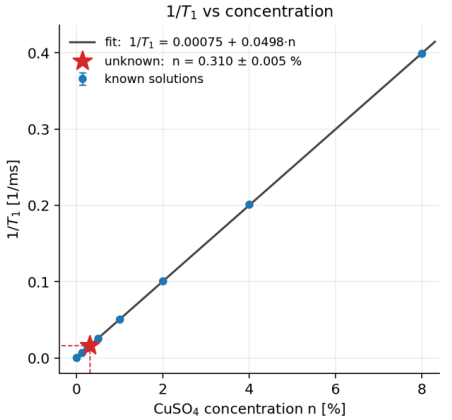
Solution	T_2 [ms]	χ^2_{red}
8%	2.060 ± 0.022 (1.06%)	122.7
4%	4.329 ± 0.074 (1.71%)	15.40
2%	8.622 ± 0.082 (0.95%)	12.07
1%	17.40 ± 0.21 (1.19%)	8.01
0.5%	34.07 ± 0.46 (1.35%)	7.01
0.25%	67.70 ± 1.22 (1.81%)	4.16
0.125%	131.2 ± 2.4 (1.85%)	12.93
H_2O	1426 ± 457 (32.0%)	1.28
Glycerine	33.32 ± 0.50 (1.49%)	9.34
Unknown	54.30 ± 0.61 (1.12%)	2.83

T_2 grows with dilution over the same three decades as T_1, and the hierarchy $T_2^* \le T_2 \le T_1$ can be confirmed for every sample. The large 32% uncertainty of the water value likely stems from the limited length of the echo train. With T_2 of order one second, repetition-time and field-drift constraints allowed us to cover only about $0.3\,T_2$. The envelope therefore shows little curvature, which loosely constrains T_2, even though the fit itself is good ($\chi^2_{\mathrm{red}} = 1.28$). The remaining fits show χ^2_{red} between 2.8 and 123, with the largest values on the strongest echo trains. This suggests that the per-point uncertainty we assigned, the additive baseline noise, is not the only source of scatter. The echo amplitudes also fluctuate multiplicatively by a few percent through pulse-width jitter and H_1 inhomogeneity, effects that scale with the signal and dominate for strong trains, inflating χ^2_{red}. The fitted T_2 values still appear reliable, since they fall on the same calibration line as the independently measured T_1 values, but where $\chi^2_{\mathrm{red}} \gg 1$ the quoted uncertainties should be treated as lower bounds.

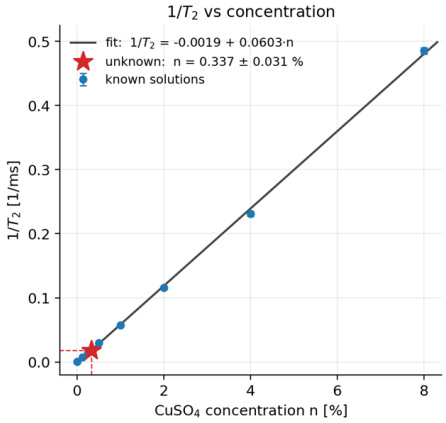
Water is a separate case. The long τ needed to span its second-scale decay allows the molecules to diffuse across the residual field inhomogeneity between refocusing pulses, adding an extra attenuation that CPMG cannot refocus [2], consistent with the measured $T_2 < T_1$. Glycerine also stands out, with a T_2 well below its own T_1, which is taken up in the discussion.

4.5. Concentration of the Unknown Solution

The rates $1/T_1$ and $1/T_2$ of the known solutions were fit to the linear calibration of Eq. (14) and the unknown sample projected onto each line. Both fits are shown in Fig. 11 and the resulting estimates in Table V.



(a) $1/T_1$ calibration.



(b) $1/T_2$ calibration.

FIG. 11: Calibration fits of the two relaxation rates versus $CuSO_4$ concentration n. Blue points are the known solutions, the line is the linear fit, and the red star marks the unknown sample.

TABLE V: Concentration of the unknown $CuSO_4$ solution from the two calibration curves. Rates are in ms^{-1} and n in %, so the slopes carry units of $ms^{-1}\,\%^{-1}$. Values in parentheses are relative uncertainties (%).

Fit	Model $b + m\,n$	n_{unknown} (%)
$1/T_1$	$0.00075 + 0.0498\,n$	0.310 ± 0.005 (1.6)
$1/T_2$	$-0.0019 + 0.0603\,n$	0.337 ± 0.031 (9.2)

The two values agree within their mutual uncertainties ($N_\sigma \approx 0.9$). Combining them by an inverse-variance weighted mean gives $n_{\mathrm{unknown}} = (0.3106 \pm 0.0049)\,\%$.

5. DISCUSSION AND CONCLUSIONS

5.1. The Gyromagnetic Ratio and the Field

All ten samples returned the same gyromagnetic ratio within uncertainty, consistent with γ being a property of the proton and not of its host liquid. The raw average disagreed with the accepted value $\gamma_p = 26754 \pm 3$ rad s^{-1} G^{-1}. The discrepancy is consistent with a single documented systematic. The magnet was 6.3 °C above its calibration temperature during the measurements, and once the corresponding drift in H_0 is applied through Eq. (17), the corrected value $\gamma_{\mathrm{corr}} = 26726 \pm 25$ rad s^{-1} G^{-1} agrees with γ_p. The same coefficient illustrates how thermally fragile a permanent-magnet spectrometer is, since a drift of a single degree moves the proton resonance by roughly 17 kHz, three orders of magnitude above the synthesiser resolution.

Beyond γ, the FID provides a direct measurement of the magnet itself. Inverting the FID width with γ_p gives a field spread across the sample of $\Delta H \approx 0.26$ G, or about 72 ppm of H_0. This is far below the 10 G resolution of the Hall probe. As shown by Lim et al. [5], precision determinations of γ_p are obtained from NMR measurements on water samples rather than from external-field references, since no field reference is sufficiently precise to match the precision of the proton itself.

5.2. Relaxation as a Concentration Probe

Both relaxation rates grow linearly with the Cu^{2+} concentration across the full range of doped solutions (0.125% to 8%), during which T_1 itself changes by nearly three orders of magnitude. This confirms the dilute-limit picture of Eq. (14), in which each ion relaxes its own neighbourhood independently and their contributions simply add. The same linearity is what allows paramagnetic contrast agents to work in magnetic resonance imaging [1], and it is what makes the unknown concentration recoverable in Sec. 4.5.

The fits also predict the undoped solvent, through the intercepts $b_i = 1/T_{i,0}$ of Eq. (14). This is the one fitted prediction we can test against a real sample, our distilled water, and on both lines it fails. The $1/T_1$ line gives $T_{1,0} \approx 1.3$ s, well short of the measured 2.004 ± 0.008 s. The $1/T_2$ line gives a negative rate, which no liquid can have, echoing the CPMG bias of Sec. 4.4. Neither failure is physical. Both intercepts extrapolate two decades below the lowest standard, from fits anchored at 1–8%, so they simply mark the edge of the calibration. Within its measured range of 0.125–8% the calibration remains sound, and the unknown, bracketed by the 0.25% and 0.5% standards, is an interpolation that is unaffected.

Our distilled water is not perfectly undoped either. Its $T_1 = 2.0$ s falls short of the ∼3 s of fully degassed water. Dissolved atmospheric O_2 is paramagnetic and makes air-equilibrated water faintly Cu^{2+}-like [1], a weak self-doping far too small to affect the unknown result.

We also find that the ions shorten T_2 slightly more than T_1. Across every doped solution the ratio T_1/T_2 stays within a narrow range of 1.12–1.22, with no trend in concentration. In a fast-moving liquid like water the two times should be equal, so this small gap points to an extra contribution to $1/T_2$ that is absent from $1/T_1$. A plausible source is a slowly varying component in the magnetic field the Cu^{2+} electron spin produces at nearby protons. The rapid fluctuations near the Larmor frequency relax T_1 by carrying energy away. The slow component cannot do this, but it can still push neighbouring spins out of step and so add to $1/T_2$ alone. This keeps T_2 below T_1, as the expected hierarchy requires. Since the gap does not change with dilution, it most likely comes from each ion acting on the protons immediately around it rather than from any collective property of the solution.

5.3. Glycerine as a Probe of Molecular Motion

TABLE VI: Ratio T_1/T_2 for each sample, formed from Tables II and IV.

Solution	T_1/T_2
8%	1.22
4%	1.15
2%	1.15
1%	1.14
0.5%	1.15
0.25%	1.14
0.125%	1.12
Unknown	1.14
Glycerine	1.64

Glycerine is the one sample in our set with no paramagnetic ions but a high viscosity, which makes it a direct probe of a second relaxation mechanism, namely the effect of molecular motion. Two features set it apart. First, despite containing no ions, it relaxes as quickly as a ∼0.35% $CuSO_4$ solution. Second, as Table VI shows, it is the only sample in which T_1 and T_2 separate strongly. Its ratio of 1.64 ± 0.03 sits far above the 1.1–1.2 band shared by every doped solution. Both features can be traced

to how fast the glycerine molecules tumble. Water is omitted from Table VI. Its ratio would be $T_1/T_2 \approx 1.41$, but the 32% uncertainty on its T_2 makes this value unreliable. The link between the ratio and the molecular motion is given by the theory of Bloembergen, Purcell and Pound [6]. If the local field at a proton loses correlation over a rotational time τ_c, with spectral density $J(\omega) = \tau_c/(1+\omega^2\tau_c^2)$, the two rates are

$$\frac{1}{T_1} = K\left[J(\omega_0) + 4J(2\omega_0)\right] \tag{25}$$

$$\frac{1}{T_2} = \frac{K}{2}\left[3J(0) + 5J(\omega_0) + 2J(2\omega_0)\right] \tag{26}$$

with K a constant of the spin pair, and $\omega_0 = \gamma H_0$ the proton Larmor frequency from Eq. (3), the rate at which the spins precess in the static field. In other words, each rate measures how much of the field's fluctuations fall within the frequencies to which the spins are sensitive. T_1 responds only to fast fluctuations near ω_0 and $2\omega_0$, which can carry energy away, while T_2 also responds to the slow, near-static fluctuations captured by the $J(0)$ term. Taking the ratio cancels K, leaving a quantity that depends only on $\omega_0\tau_c$. For fast motion ($\omega_0\tau_c \ll 1$) the two rates coincide and $T_1 = T_2$, whereas slower motion enlarges $1/T_2$ through the zero-frequency term $J(0)$ and opens the gap. Inverting the measured ratio and performing a numerical calculation gives $\omega_0\tau_c = 0.64 \pm 0.02$, or a correlation time $\tau_c = 6.7 \pm 0.2$ ns. Two macroscopic decay times therefore yield the nanosecond timescale on which individual glycerine molecules reorient. This value also resolves why an ion-free liquid relaxes so quickly. Relaxation is most efficient near $\omega_0\tau_c \approx 0.6$, the so-called T_1 minimum [1], and glycerine at room temperature falls almost exactly there. Its rapid relaxation is thus a consequence of molecular tumbling landing near this optimum, not of any paramagnetic content. The contrast with the unknown sample makes the distinction concrete. The two have nearly equal T_1 (Table II), yet their ratios differ sharply (1.14 against 1.64), which identifies paramagnetic doping in one and slow molecular motion in the other. A single relaxation time fixes the rate of relaxation, but only the pair (T_1, T_2) distinguishes its cause.

5.4. The Unknown Concentration

Combining everything measured, the unknown was characterised without any chemical analysis. Its $T_1 = 61.7$ ms is close to that of glycerine (Table II), so a single relaxation time cannot say whether it is a lightly doped ionic solution or a viscous, ion-free liquid like glycerine. The ratio is what separates the two possibilities. At $T_1/T_2 = 1.14$ the unknown sits in the $CuSO_4$ band of Table VI rather than near glycerine's 1.64, which suggests that its fast relaxation comes from paramagnetic ions rather than from slow molecular tumbling.

If this is correct, the unknown is most likely a dilute $CuSO_4$ solution, consistent with the calibration set it was measured against. Projecting it onto the calibration would then fix the amount at $n = (0.311 \pm 0.005)\%$, an interpolation between the 0.25% and 0.5% standards. Arrived at entirely from the relaxation behaviour of protons, this identification draws on all three measured time constants: T_2^* established the homogeneity of the field, the pair (T_1, T_2) pointed to an ionic dopant rather than a viscous liquid, and the rates $1/T_1$ and $1/T_2$ then quantified how much of that dopant was present.

[1] R. L. Dixon and K. E. Ekstrand, "The physics of proton NMR," *Medical Physics*, vol. 9, no. 6, pp. 807–818, 1982.

[2] H. Y. Carr and E. M. Purcell, "Effects of diffusion on free precession in nuclear magnetic resonance experiments," *Physical Review*, vol. 94, no. 3, pp. 630–638, 1954.

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[4] TeachSpin, Inc., Buffalo, NY, *PSA-1 Pulsed NMR Spectrometer: The Instrument*, 1994.

[5] A. R. Lim, C. S. Kim, and S. H. Choh, "Effect of paramagnetic ions in aqueous solution for precision measurement of proton gyromagnetic ratio," *Bulletin of Magnetic Resonance*, vol. 14, pp. 240–245, 1992.

[6] E. R. Andrew, *Nuclear Magnetic Resonance*. Cambridge: Cambridge University Press, 1955.

Appendix A: Uncertainty Assignment and Propagation

All measurement uncertainties quoted in this report trace back to one of the sources below.

1. Instrument-Resolution Readings

Values read from a digital instrument (the 10 G Hall probe) were treated as uniformly distributed over the resolution window, giving a standard uncertainty of the resolution divided by $\sqrt{12}$.

2. Quantities Read Repeatedly

Values obtained from several independent readings that fluctuate between firings (the Larmor frequency and the inversion-recovery peak amplitudes) were assigned the centre of the observed spread as the value and half the spread as the uncertainty.

3. Points of a Single Digitised Trace

Points sampled from one digitised trace (the FID envelope and the CPMG echo peaks) were assigned a uniform per-point uncertainty equal to the RMS noise of the prepulse baseline of that trace.

4. Fitted Parameters

All model fitting was performed offline by weighted least squares. Fitted parameters carry the 1σ standard errors of the least-squares covariance matrix, and derived quantities follow from first-order propagation, retaining parameter covariances where relevant.

5. Propagation for the Unknown Concentration

The concentration estimate of Eq. (22) depends on the measured rate $y = 1/T_{i,\mathrm{unknown}}$ and on the fitted parameters (b_i, m_i), which are correlated through the fit. First-order propagation gives

$$\sigma_n^2 = \frac{1}{m^2}\left[\sigma_y^2 + \sigma_b^2 + n^2\sigma_m^2 + 2n\,\mathrm{cov}(b,m)\right] \tag{A1}$$

where σ_y follows from the fitted relaxation time, σ_b and σ_m are the standard errors of the calibration fit, and $\mathrm{cov}(b,m)$ is taken from its covariance matrix. The two estimates $n_i \pm \sigma_{n_i}$ were combined by the inverse-variance weighted mean

$$\bar{n} = \frac{\sum_i n_i/\sigma_{n_i}^2}{\sum_i 1/\sigma_{n_i}^2} \qquad \sigma_{\bar{n}} = \left(\sum_i \frac{1}{\sigma_{n_i}^2}\right)^{-1/2} \tag{A2}$$

which yields the value quoted in Sec. 4.5.